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MGSC 670- Revenue Management

Revenue Management

Case 2:Cargo

Please note all supporting calculations and support are available in the accompanying excel document.

Part 2: Cargo and its Suppliers

1 We consider and simulate the supply chain under a wholesale price contract. As mentioned, we use a retail price of $10, a unit production cost of $2.50, and a (monthly) demand that is normally distributed with mean 1,000 and standard deviation 200. The demand realizations are given in the spreadsheet.

1.1 Under a wholesale price of $5, what is the retailer's optimal order quantity?

Based on these conditions the retailer's **optimal order quantity is 1,000 units**. This can be found using the NORMIV function in Excel with the following logic:
NORMINV(Critical_val,average,stdev) or NORMINV(.5,1000,200)

- W=$5
- P=$10
- C=$2.5
- Co=w=$5
- Cu=p-w=$10-$5=$5
- Critical fractile = Cu / (Cu+Co) = 0.5

1.2 Under a wholesale price of $5, compute the expected profit of the retailer and of the supplier.

Based on a cost c = $2.5, a wholesale price w = $5 and a retail price p = $10. Cu of the cost of understocking is $5 or the wholesale price, while the cost of overstocking Co is $10-$5 = $5. The average sum of the retailer or **the drivers is $4,247.81** while the average sum of the supplier **Cargo is $2,500.00.** Therefore the **total supply chain sum is $6,747.81**

1.3 Vary the value of the wholesale price between $2.50 and $10 and find the value that yields the highest possible profit for the supplier. What is this wholesale price value?

By varying the cost of the wholesale price w between $2.50 and $10 the optimal value that yields the highest possible **profit for the supplier is when w=$9.00**. When w=$10, the critical fractal is undefined, as mathematics and logic do not work. The average profit for the supplier, in this case, is **$4,833.99.**

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1.4 For the wholesale price value obtained in Part 1c, what is the total expected profit of the supply chain (i.e., the sum of the retailer's profit and the supplier's profit)?

For a wholesale price of $9.00 there is a **total supply chain profit of $5,487.45.** The retailer's profit is **$656.31,** while the supplier's is **$4,833.98 .**

2 The supply chain's First-Best profit is defined as the profit under the ideal situation where the entire supply chain is owned by one party, i.e., the centralized supply chain.

2.1 What is the supply chain's optimal order quantity?

The retailer's **optimal order quantity is 1,135 units** based on these conditions. More exactly 1,134.89 units. This can be found using the NORMIV function in Excel with the following logic: NORMINV(Critical Val, average, stdev) or NORMINV(.75,1000,200)

- W=$5
- P=$10
- C=$2.5
- Co=c=$2.5
- Cu=p-w=$10-$2.5=$7.5
- Critical fractal = Cu / (Cu+Co) = 0.75

2.2 What is the First-Best expected profit?

Based on the optimal order quantity of 1,135 the **first best-expected profit is $6,964.55.** with $4,127.30 from the retailer and $2,836.24 from the supplier.

2.3 How does the retailer's order quantity (from Part 1a) compared to the First-Best's optimal order quantity (from Part 2a)? Why is it the case? Carefully justify your answer.

The optimal order quantity from question 7.1 is 1,000 versus, 1,135 for question 8.1. The order quantity for question 8.1 is much higher as the critical fractile or ratio ofthe cost of overstocking and understocking is much higher. This therefore indicates that the supply chain overall is incentivized to overstock on items since the cost of overstocking is less than the penalty of understocking.

In other words when the cost of understocking is equal to the cost of overstocking, the math estimates a lower inventory allocation to optimize the amount of inventory as close to the mean as possible. There is an equal chance of over or under demand in a normal distribution, therefore the math determines the approach is to be equally wrong in terms of over and understocking. Any demand above the inventory amount is not met, and any demand under the inventory is not met but both cost the same.

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If overstocking costs are lower than understocking, the supply chain is better off overstocking to capture more of the demand above the average demand. The supply chain can easily capture demand above the average and gain additional revenue at low cost. Whereas it would cost 3 times as much in cost or potential revenue to not meet demand. In other words, the cost of overstocking and understocking is balanced with a higher inventory number given a known average and standard deviation.

By lowering the price of inventory we can increase the inventory on hand and meet more demand.

2.4 How does the total expected profit (from Part 1d) compare to the First-Best expected profit (from Part 2b)? What is the relative difference between these two values?

The total supply chain profit from part 7.4 is $5,487.45, while the first best-expected profit is $6,964.55. Therefore, the first-best profit **when the supply chain is centralized the profit is higher by $1,477.09,** a centralized approach is more profitable based on these assumptions and conditions.

3 (Optional) Re-solve questions 1 and 2 using a salvage value of $1. How does it affect the results? Please elaborate.

For question 3 we will assume that the retailer receives 100% of the salvage value.

3.1 Question 1

3.1.1 Under the wholesale price of $5, what is the retailer's optimal order quantity?

The optimal order quantity is **1,028 units.** More exactly 1,027.98 units.

NORMINV(Critical_val,average,stdev) or NORMINV(.55,1000,200)

- W=$5
- P=$10
- S=1
- C=$2.5
- Co=w-s=$4
- Cu=p-w=$10-$5=$5
- Critical fractile = Cu / (Cu+Co) = 0.555

3.1.2 Under the wholesale price of $5, compute the expected profit of a retailer and of the supplier

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The expected profit of the **retailer is $ $4,322.13** while that of the **supplier is $ $2,569.86 . The total supply chain profit is $6,891.99.**

3.1.3 Vary the value of the wholesale price between $2.50 and $10 and find the value that yields the highest possible profit for the supplier. What is this wholesale price value?

The price that yields the highest possible profit for the supplier is **w=8.99**, this change is likely due to the the salvage value. The maximum supplier profit is **$4,913.20.**

3.1.4 For the wholesale price value obtained in Part 1c, what is the total expected profit of the supply chain (i.e., the sum of the retailer's profit and the supplier's profit)?

The total expected profit of the supply chain based on part 9.1.3 **is $ $5,842.80** The **retailer's average profit is $929.61** while the **supplier's is $ $4,913.20**

3.2 Question 2

3.2.1 What is the supply chain's optimal order quantity?

The optimal order quantity is **1,194 units.** More exactly 1,193.48 units

NORMINV(Critical_val,average,stdev) or NORMINV(.55,1000,200)

- W=$5
- S=1
- P=$10
- C=$2.5
- Co=c-s=$1.5
- Cu=p-c=$10-$2.5=$7.5
- Critical fractile = Cu / (Cu+Co) = 0.833

3.2.2 What is the First-Best expected profit?

The first-best expected **profit is $6,973.56.**

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3.2.3 How does the retailer's order quantity (from Part 1a) compared to the First-Best's optimal order quantity (from Part 2a)? Why is it the case? Carefully justify your answer.

The optimal order quantity from question 3.1.1 is 1,028 versus 1,194 for question 3.2.1. The order quantity for the first-best solution has increased as the supply cost is much lower in this model than in the wholesale model, meaning that the cost of overstocking has reduced between these two models.

3.3 Salvage value effects

Both models increased their profitability by re-running the models with salvage costs. This makes sense as the salvage value decreases the oversupply cost. Since there is some cost recovery from unsold items, the supplier and retailer can acquire more inventory at no risk in a centralized model. This thus leads to a decision to carry a higher inventory for the retailer and supplier in the decentralized model. In the decentralized model, the primary source of increased profitability is the supplier as the retailer's profits remain stable, the additional costs offset the additional demand.

In the first-expected profit, the overall supply chain's profit is maximized through the reduced cost of goods sold. The number of units in inventory increases and is higher than the amount without a salvage value of 1,135 from question 2.1. The supplier profit decreases while that of the retail increases. Even though the supplier and retailer's total profit is maximized, and the salvage value increases the profit of both parties, the centralized model is a clear win for the retailer.

	W=5	**W=8.99**	**Centralized**
w	$5	$8.99	$5
c	$2.5	$2.5	$2.5
p	$10	$10	$10
s	$1	$1	$1
Co	w-s= $4	w-s= $7.99	c-s= $1.5
Cu	p-w=$5	p-w=$1.01	p-c=$7.5
Critical Fractal	0.55	0.11	0.83
Optimal Order Quantity	1028	757	1193
Expected Supplier Profit	$4,247.32	$4,913.20	
Expected Retail Profit	$2,569.86	$929.61	
Total Expected Profit	$6,817.17	$5,842.80	$6,973.36

4 We now consider using a revenue sharing contract. We start with a wholesale price w of $0.75 and a revenue share percentage y of 0.7.

4.1 What is the retailer's optimal order quantity? What can you say on this value?

The retailer's **optimal order quantity is 1,135 units** based on these conditions. This can be found using the NORMIV function in Excel with the following logic: NORMINV(Critical Val, average, stdev) or NORMINV(.75,1000,200)

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- w=$0.75
- P=$10
- C=$2.5
- Co=w=$0.75
- Cu=p(1-y)-w=$10*(1-0.7)-$0.75=$2.25
- Critical fractal = Cu / (Cu+Co) = 0.75

4.2 Compute the retailer's expected profit and the supplier's expected profit under the above revenue sharing contract. What can you conclude?

The **retailer's expected profit is $2,089.36,** while the **suppliers is $4,875.18. The total supply chain profit is $6,964.55** under the revenue-sharing contract. Therefore, the profit is larger for both than under the wholesale contract. This method also creates more equity and motivation throughout the supply chain. However, the supplier does benefit from a larger profit when the first-best expected profit method is used, where the supplier maximizes the wholesale price of the item.

4.3 When comparing the wholesale price contract to the revenue sharing contract, who benefits? Justify your answer

Profit	Wholesale Price Contract	Revenue Sharing
Supplier	$4,834	$4,875
Retailer	$656	$2,089
Total	$5,490	$6,964

The revenue-sharing agreement generates more profit based on the profits of the wholesale and revenue-sharing contracts. Not only does the profitability increase overall, but for both the suppliers are retailers. Therefore, **both the supplier and retailer are strictly better off and benefit** under the revenue-sharing contract compared to the wholesale agreement.

An adequately designed supply chain can empower the supplier and retailer by making them both stakeholders while taking advantage of the pricing and risks to generate the same profit as a "greedy" first-best profit that only benefits the supplier. A supply chain's maximum utility or profitability is under the revenue-sharing methodology.

5 In Part 4b, you calculated the expected profit earned by the supplier and by the retailer. In some settings, to incentivize the retailer to sign on, the supplier needs to guarantee a minimum expected profit level to the retailer. Find the wholesale price w and the revenue share percentage y that maximize the supplier's expected profit while ensuring the retailer makes at least as much as under the wholesale price contract (see Part 1b).

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The retailer's profit from question 1.3 is $4,247.81**.** We will vary the wholesale price w between $0.01 and $9.99 as these are the price parameters that make the most sense given the circumstances. Additionally, the split will be allowed to vary between 0.01 and 0.99 initially to capture the optimal result best. The sharing percentage cannot be 0 or 1 as this would eliminate the purpose of revenue sharing entirely.

After varying the wholesale price w and the revenue sharing profit % we find that a wholesale price of **w=$1.33 with a % sharing of y=41.30%** for the supplier meets the criterial. The profit generated for the retailer is $4,247.81 while the supplier generates $2,719.82. Therefore there is a total of **$6,967.62 in supply chain profit.**

6 In a buyback contract, the supplier does not receive a share of the retailer's revenue. Instead, the supplier offers to buy back any unsold unit from the retailer (at a discount) at the end of the selling period. Implement such a contract in this context. What is the optimal buyback price? How does this contract perform relative to the wholesale price contract and relative to the revenue sharing contract?

Since we are basing the evaluation on the wholesale price contract, we will use certain assumptions from this model. The buyback price will be considered a salvage value and mathematically be a percentage multiplied by the wholesale price. We will also need to reduce the supplier's profit by the buyback amount since they are spending money to acquire the excess stock.

6.1 Win-win Strategy

We will also assume that the optimal solution is one where the revenue for the supply chain is the greatest, and the profit for both the suppliers and the retailers is higher than in the wholesale situation. Therefore, the optimal buyback percentage is **0% of the wholesale cost**. In this case, there is no percentage the will allow for a win-win compared to the wholesale method and allow for a win-win. The **retailer's profit is $4,247.81**, the same as the wholesale scheme. While that of the **supplier is $2,500, and the total supply chain profit is $6,747.81**

- w=$5
- P=$10
- C=$2.5
- S=%buyback * w =$5*100%=$5
- Co=w-s=$4.09
- Cu=p-w=$5
- Critical fractal = Cu / (Cu+Co) = 0.55
- Order quantity=1000
- Discount percent for buy back=0%

6.2 No Win-win Strategy

If we relax the assumption that the retailers and suppliers need higher or equal profits, assuming instead maximizing supply chain. We seek to maximize the supply chain profit, we get a very different outcome. The optimal buyback percentage is **70.69% of the wholesale cost or $3.53**. In this case, the

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retailer's profit is $4,678.48, lower than in the wholesale scheme. While that of the **supplier is $2,289.15** higher than in the wholesale conditions**, and the total supply chain profit is $6,967.62.**

- w=$5
- P=$10
- C=$2.5
- S=%buyback * w =$5*0.2619=3.53
- Co=w-s=$1.47
- Cu=p-w=$5
- Critical fractal = Cu / (Cu+Co) = 0.7733
- Inventory quality=1,150
- Discount percent for buy back=100-26.16=29.31%